CHAPTER – 4
ANALYZING FUNCTIONAL DEPENDENCIES, REDUNDANCY AND DATA ANOMALIES

INTRODUCTION

The **Tira Online Beauty & Cosmetics Shopping Database System** is designed to efficiently manage customers, beauty products, categories, shopping carts, orders, payments, deliveries, reviews, and administrators. Functional dependencies identify the relationship between attributes in each table, while redundancy analysis and data anomaly detection ensure data consistency and reduce data duplication. A well-designed database improves performance, maintains data integrity, and supports efficient online shopping.

1. CUSTOMER TABLE

Functional Dependency

Customer_ID → First_Name, Last_Name, Email, Phone, Date_of_Birth, Gender, Address

Redundancy Analysis

Customer information is stored only once in the Customer table. Orders store only the **Customer_ID**, preventing duplicate customer records.

Data Anomalies

- **Insert Anomaly:** A new customer can be added without placing an order.
- **Update Anomaly:** Updating a customer's details requires changes only in the Customer table.
- **Delete Anomaly:** Deleting an order will not remove customer information.

2. PRODUCT TABLE

Functional Dependency

Product_ID → Product_Name, Brand_ID, Category_ID, Price, Shade, Stock

Redundancy Analysis

Product details are stored only in the Product table. Orders reference products using **Product_ID**, avoiding repeated product information.

Data Anomalies

- **Insert Anomaly:** New products can be added before any order is placed.
- **Update Anomaly:** Product price or stock is updated only once.
- **Delete Anomaly:** Deleting an order does not remove product information.

3. ORDER TABLE

Functional Dependency

Order_ID → Customer_ID, Order_Date, Order_Status, Total_Amount

Redundancy Analysis

The Order table stores only order-related information. Customer details are referenced using **Customer_ID**.

Data Anomalies

- **Insert Anomaly:** Orders can be created for registered customers.
- **Update Anomaly:** Updating the order status affects only the Order table.
- **Delete Anomaly:** Deleting an order does not remove customer or product details.

4. CART TABLE

Functional Dependency

Cart_ID → Customer_ID, Total_Items, Grand_Total

Redundancy Analysis

Each customer has a separate cart. Product details are not duplicated in the Cart table.

Data Anomalies

- **Insert Anomaly:** A cart can be created before adding products.
- **Update Anomaly:** Cart total is updated only in one place.
- **Delete Anomaly:** Deleting a cart does not delete customer information.

5. ORDER_ITEM TABLE

Functional Dependency

OrderItem_ID → Order_ID, Product_ID, Quantity, Unit_Price

Redundancy Analysis

The Order_Item table connects Orders and Products without repeating product information.

Data Anomalies

- **Insert Anomaly:** Products can be added to an order independently.
- **Update Anomaly:** Quantity changes affect only the selected order item.
- **Delete Anomaly:** Removing an order item does not delete the product.

6. DELIVERY TABLE

Functional Dependency

Delivery_ID → Order_ID, Delivery_Address, Delivery_Date, Delivery_Status

Redundancy Analysis

Delivery information is stored separately to avoid repeating delivery details.

Data Anomalies

- **Insert Anomaly:** Delivery details can be added after order confirmation.
- **Update Anomaly:** Delivery status is updated only once.
- **Delete Anomaly:** Deleting delivery information does not remove the order.

7. ADMINISTRATION TABLE

Functional Dependency

Admin_ID → Admin_Name, Email, Password

Redundancy Analysis

Administrator details are stored separately to avoid duplication.

Data Anomalies

- **Insert Anomaly:** New administrators can be added independently.
- **Update Anomaly:** Admin details are updated only once.
- **Delete Anomaly:** Deleting an administrator does not affect customer or order records.

8. REVIEW TABLE

Functional Dependency

Review_ID → Customer_ID, Product_ID, Rating, Review_Text, Review_Date

Redundancy Analysis

Reviews are stored separately, preventing repeated customer feedback.

Data Anomalies

- **Insert Anomaly:** A review can be added after purchasing a product.
- **Update Anomaly:** Review content can be updated without affecting product details.

- **Delete Anomaly:** Deleting a review does not remove customer or product information.

9. CATEGORY TABLE

Functional Dependency

Category_ID → Category_Name, Description

Redundancy Analysis

Each category is stored only once. Products reference the category using **Category_ID**.

Data Anomalies

- **Insert Anomaly:** New categories can be created without products.
- **Update Anomaly:** Category name is updated only once.
- **Delete Anomaly:** Deleting a product does not remove the category.

10. PAYMENT TABLE

Functional Dependency

Payment_ID → Order_ID, Payment_Method, Payment_Status, Payment_Date, Amount

Redundancy Analysis

Payment details are stored separately from the Order table to avoid duplicate payment records.

Data Anomalies

- **Insert Anomaly:** Payment can be recorded after an order is created.
- **Update Anomaly:** Payment status is updated only once.
- **Delete Anomaly:** Deleting payment information does not remove the order.

CONCLUSION

The **Tira Online Beauty & Cosmetics Shopping Database System** is designed using proper functional dependencies and normalized tables. Each entity stores only its relevant data, reducing redundancy and preventing insert, update, and delete anomalies. This design improves data consistency, maintains data integrity, and ensures efficient database management for the Tira online shopping system.